D07 · Warehouse KPIs, Metrics & Governance

02 · Space utilization & capacity planning

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1 WHAT IT IS

Space utilization measures how much of the warehouse's positions and volume are actually occupied; capacity planning projects when that space and the site's throughput will run out, so expansion or densification starts in time.

2 WHY IT MATTERS

A warehouse running near full has no room to maneuver: congestion rises, slotting degrades and every peak becomes a crisis. **A warehouse far below capacity pays for space it does not use.** The useful number is not today's utilization but the months of capacity remaining at the current growth rate.

3 HOW IT IS DONE

1. **Measure position and volume utilization.** Including height usage, by zone.
2. **Project remaining capacity.** Months until the operational limit, from the inventory and volume plan.
3. **Free space before building it.** SLOB reduction, re-slotting and densification options come first.
4. **Plan expansion through S&OP.** Trigger decisions with enough lead time to act without urgency.

IN PRACTICE

Manage warehouse space as a financial asset — not an operational constraint

m² has a real cost. Measuring cost per occupied m² and cost per utilized pallet position connects space management with the logistics P&L.

The SLOB reduction program is the fastest way to release capacity without investment

SLOB occupies warehouse positions without generating value. Its elimination releases space without infrastructure investment.

Plan warehouse capacity with the S&OP plan — not reactively

If projected volume growth for the next 12 months exceeds available capacity, expansion must start today — not when the warehouse is full.

Evaluate warehouse densification before footprint expansion

Higher-density shelving (drive-in, push-back, pallet shuttle) and increasing existing rack height can increase capacity 20–40% without footprint expansion.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Capacity planning — DC in space crisis, 94% utilization at Q4 peak — Operations are degrading: congestion, slotting errors, overtime.

Capacity planning action	Timeline	Utilization impact
SLOB reduction program (120 positions freed)	1 month	Utilization: 94% -> 88% · SLOB eliminated: $480K of low-velocity inventory
Re-slotting to increase A and B SKU density	2 months	180 additional positions without construction · Effective utilization: 85%
Conversion of 2 receiving docks to temporary Q4 peak storage	Seasonal	Additional capacity: +240 positions during 6 peak weeks

Result: Q4 capacity crisis resolved with the first 3 actions (no infrastructure investment). ASRS evaluation for the medium term initiated — the 15%/year growth trend makes investment inevitable in the next 3 years.

HOW IT IS MEASURED

Warehouse Space Utilization %

Warehouse Space Utilization %Occupied pallet positions / Total available positions × 100

Benchmark: 70–85% as operational target · >90% generates congestion · <60% indicates excessive space being paid for without use A warehouse consistently >90% utilization is a warehouse with no room to maneuver. The next demand peak will generate customer-visible operational problems.

Months of Remaining Capacity (months until filled to 90%)

Months of Remaining Capacity(0.90 × Total capacity - Current inventory) / Average monthly inventory increment

Benchmark: >6 months remaining capacity to plan expansion without urgency · <3 months requires immediate action Remaining capacity horizon is the most strategic space management indicator. A <3 month horizon means the space crisis is already underway.

WHAT YOU WOULD USE

Capacity Planning

SAP EWM — Warehouse Management Analytics — Module: Space Utilization + Capacity ReportingSAP EWM reports warehouse position utilization and projects remaining capacity based on the inventory plan.

Power BI — Warehouse Capacity Dashboard — Module: Space Utilization + Capacity ForecastingPower BI dashboard for space utilization tracking with remaining capacity projection.

4 SO WHAT

Manage space as a financial asset. Track utilization by zone, project remaining months, free space through SLOB and densification, and take expansion decisions in S&OP well before the peak.

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